

Doc. 309 – The Scale-Anchor Problem: Λ CDM and FFGFT Compared

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Contents

Doc. 309 – The Scale-Anchor Problem	2
A The SI System — a Network without a First Anchor	3
A.1 All units traceable to one	3
A.2 There is no first anchor	3
B Λ CDM and its Scale Anchor	4
B.1 Six free parameters, all empirical	4
B.2 H_0 : measured circularly	4
B.3 The fine-tuning problem	4
C FFGFT and its Scale Anchor	5
C.1 Structure of ξ [K] and magnitude [POSIT]	5
C.2 The actual fit: the exponent 10	5
C.3 The inheritance: FFGFT inherits Λ CDM systematics [K]	5
C.4 Cross-anchoring: what FFGFT achieves nonetheless	6
D Comparison	7
E Open Points (Doc. 190)	8

Doc. 309 – The Scale-Anchor Problem

What this is about

Both cosmological theories — Λ CDM and FFGFT — must introduce somewhere a number that couples the cosmic scale to the SI system. Neither derives this anchor from first principles.

This is not a coincidence — it is a structural problem of every physical theory that has dimensions. But the **kind and quality** of the anchor differ fundamentally. And there is an asymmetry that has not been explicitly named in any previous document: FFGFT inherits the systematics of the expansion reading through its scale fit.

This document is not part of the A-series. References: Doc. 190 (P20, P33), Doc. 267 (degeneracy), Doc. 279 (H_0 chain), Doc. 308 (cosmic sector).

Conventions

Legend.

- [K]** *Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
- [S]** *Structural claim* — justified but not yet fully proved.
- [POSIT]** *Posit* — declared assumption, not further derived.

Central quantities.

- $\xi = 4/30000$ Only fundamental parameter of FFGFT [POSIT, P33]. Form $(4/3, 3, 2^2)$ forward-derived [K]; magnitude (5^4) intuitive/empirical [POSIT].
- $\bar{\lambda}_e = \hbar/(m_e c)$ Reduced Compton wavelength of the electron; $= 3.8616 \times 10^{-13}$ m.
- H_0 Cosmological parameter; in FFGFT: $H_0 = (\pi/2) c \xi^{10} / \bar{\lambda}_e = 66.82$ km/s/Mpc [K], exponent 10 [POSIT, P20].
- $R_H = c/H_0$ Geometric maximal scale of the compact geometry — not an expansion horizon (Doc. 182).
- ν_{Cs} Caesium hyperfine transition; $= 9.192\,631\,770 \times 10^9$ Hz (exact, definition); conventional SI anchor.
- $\nu_e = m_e c^2 / h$ Electron frequency; $= 1.235 \times 10^{20}$ Hz; reading-independent reference frequency.
- n Integer exponent in $H_0 \propto \xi^n / \bar{\lambda}_e$; $n = 10$ is a posit (P20, Doc. 190).
- Ω_Λ Dark-energy density in Λ CDM; without reading-independent referent in the static reading.

Chapter A

The SI System — a Network without a First Anchor

A.1 All units traceable to one

Since 2019 all SI base units are defined via fixed natural constants. The seemingly “four independent measured values” c , h , e , k_B are physically not independent — they are all traceable to one single historically conventional anchor: the frequency of the caesium hyperfine transition ν_{Cs} .

Unit	Traceability	Historical origin
Metre	c/ν_{Cs} (numerical factor)	Earth meridian
Kilogram	$h \nu_{\text{Cs}}/c^2$	Prototype kilogram
Ampere	$e \nu_{\text{Cs}}$	Force definition
Kelvin	$h \nu_{\text{Cs}}/k_B$	Thermometry

A.2 There is no first anchor

ν_{Cs} was chosen in 1967 because it was technically stable and realisable — not because it is physically distinguished. The Rb transition, the electron frequency, or a hundred other systems would have served equally. All present SI numerical values would then be different — all dimensionless ratios (m_e/m_p etc.) would remain identical.

What is truly fixed: only dimensionless ratios such as $m_e/m_p \approx 1/1836$. These are unit-independent.

Consequence: Every theory that predicts cosmological absolute values in SI implicitly states a relation between two frequencies — and must set some number to do so. The FFGFT statement

$$H_0 = \frac{\pi}{2} \frac{c \xi^{10}}{\bar{\lambda}_e}$$

is equivalent to the dimensionless statement:

$$\frac{H_0}{\nu_e} = \pi^2 \xi^{10} = 1.75 \times 10^{-38} \quad \text{with} \quad \nu_e = \frac{m_e c^2}{h}.$$

Here $c/\bar{\lambda}_e = m_e c^2/\hbar = 2\pi\nu_e$; equivalently, $H_0/\omega_e = (\pi/2) \xi^{10} = 2.79 \times 10^{-39}$ with $\omega_e = m_e c^2/\hbar$.

The SI reference is contained entirely in the choice of $\bar{\lambda}_e$ as the reference scale. Without that choice, H_0/ν_e is a pure number.

Chapter B

Λ CDM and its Scale Anchor

B.1 Six free parameters, all empirical

Λ CDM describes cosmology with six free parameters (Planck 2018): H_0 , $\Omega_b h^2$, $\Omega_{\text{DM}} h^2$, n_s , σ_8 , τ . All from observation — none derived from theory.

B.2 H_0 : measured circularly

H_0 is determined via a four-rung distance ladder: parallax $\rightarrow$ Cepheids $\rightarrow$ SNe Ia $\rightarrow$ Hubble law. Each rung introduces its own calibration assumptions. The **Hubble tension** (Planck: 67.4, SHOES: 73.0 km/s/Mpc, 5σ) is a direct consequence of this circularity.

B.3 The fine-tuning problem

Λ corresponds to a vacuum energy density $\rho_\Lambda \approx 5.96 \times 10^{-27}$ kg/m³. QFT predicts $\rho_{\text{QFT}} \sim 5 \times 10^{96}$ kg/m³. The ratio is 10^{123} . Λ CDM has no explanation — it sets Λ to the observed value without theoretical origin.

Chapter C

FFGFT and its Scale Anchor

C.1 Structure of ξ [K] and magnitude [POSIT]

$\xi = 4/30000 = 1/(3 \cdot 2^2 \cdot 5^4)$. According to P33 (Doc. 190):

- **Form** ($4/3$, factor 3, factor 2^2): forward-derived [K]
- **Magnitude** ($5^4 = 625$): intuitive/empirical [POSIT]

C.2 The actual fit: the exponent 10

ξ itself comes from the geometry. The **exponent 10** in $H_0 = (\pi/2) c \xi^{10} / \bar{\lambda}_e$ was chosen so that $H_0 \approx 67$ km/s/Mpc. Since $\xi \sim 1.3 \times 10^{-4}$, H_0 jumps between neighbouring exponents by a factor of ~ 7500 :

n	H_0	Assessment
9	5.0×10^5 km/s/Mpc	unphysical
10	66.8 km/s/Mpc	plausible
11	8.9×10^{-3} km/s/Mpc	unphysical

There is no continuous fitting — $n = 10$ is the only integer choice that yields a physically sensible value. The justification remains: $n = 10$ gives a value compatible with Λ CDM measurements (P20, Doc. 190).

C.3 The inheritance: FFGFT inherits Λ CDM systematics [K]

67.4 km/s/Mpc is a Λ CDM measurement value — determined under the assumption of the expansion reading. Since $n = 10$ was calibrated to this value, FFGFT inherits the systematics of that reading.

The “natural” H_0 from the FFGFT redshift formula $z = e^{\xi x / \bar{\lambda}_e} - 1$ would be:

$$H_0^{\text{natural}} = \frac{c \xi}{\bar{\lambda}_e} \approx 3 \times 10^{36} \text{ km/s/Mpc} \quad (\text{unphysical})$$

($c \xi / \bar{\lambda}_e = 1.035 \times 10^{17} \text{ s}^{-1} = 3.19 \times 10^{36} \text{ km/s/Mpc}$; cross-check: $(\pi/2) \xi^9 \cdot 3.19 \times 10^{36} = 66.8$.)

To obtain ~ 67 km/s/Mpc, ξ^{10} is needed instead of ξ^1 — and the justification of this exponent is ultimately Λ CDM compatibility.

This is not a weakness of FFGFT alone — it is the unavoidable consequence of the degeneracy (Doc. 267): as long as both readings are empirically indistinguishable, every theory calibrates its parameters to the same data.

The remedy would be a reading-independent derivation of exponent 10 from the T^4 geometry forward — not yet achieved (P20, Doc. 190).

C.4 Cross-anchoring: what FFGFT achieves nonetheless

That ξ^{10} reproduces not only H_0 but also a_0 (inertia transition scale, Doc. 308) — that would not be expected from a pure H_0 fit. That is the cross-anchoring.

Sector	Quantity	ξ -dependence	Status
Cosmology	H_0	ξ^{10} (exponent calibrated)	[POSIT]
Inertia regime	a_0	ξ^{10}	[K]/[S]
Particle physics	lepton mass ladder	ξ -powers	[K]
Quantum mechanics	$K_{\text{frak}} = 1 - 100\xi$	ξ	[K]
Geometry	$k^* = \ln \varphi/\xi$	ξ	[S]

Ω_Λ in Λ CDM has **zero** such cross-anchorings.

Chapter D

Comparison

Property	Λ CDM	FFGFT
Free scale anchors	6–7, empirical	1 (ξ , P33)
Theoretical origin	none	none
Cross-anchoring	no	yes (5+ sectors)
Circularity	yes (H_0 ladder)	no
Inheritance of Λ CDM systematics	inherent	via exponent 10
Fine-tuning	10^{123} (Λ)	absent structurally
Hubble tension	5σ , open	not modelable

FFGFT exchanges six isolated parameters for one cross-anchored — no proof, but a reduction with direction. The inheritance of the Λ CDM scale value via exponent 10 remains the central open weakness.

Chapter E

Open Points (Doc. 190)

- **P20:** Form of H_0 as ξ -ratio fixed; exponent 10 is external calibration from Λ CDM, not derived.
- **P33:** Magnitude of ξ (factor 5^4) intuitive/ empirical, no forward proof. Structural parallel to P20.
- **Inheritance (new, Doc. 309):** Exponent 10 calibrated to $H_0^{\Lambda\text{CDM}}$. Remedy: reading-independent derivation from T^4 geometry — not yet achieved.

On the Hubble tension

The Hubble tension (5σ between early-universe calibration from CMB/BAO and the local distance ladder) is *not modelable* within FFGFT, not model-independently absent. FFGFT treats H_0 not from an expanding geometry but as an emergent, decompactified quantity, and therefore knows *only one* H_0 value (the calibrated one), without distinguishing between early- and late-time measurements. This is a weakness, not a strength: FFGFT can neither reproduce nor resolve the tension, because the geometry contains no expanding causal structure from which two different measurement-epoch effects could follow. The entry “not modelable” in the comparison table accordingly means “not provided for in the model”, not an explanatory claim.